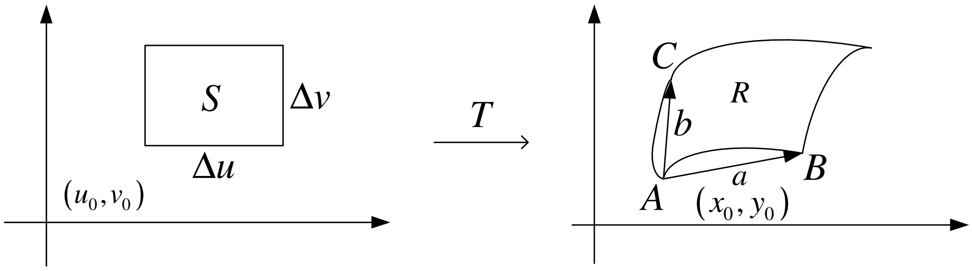
§15.9 Change of Variables in Multiple Integrals

Homework： 3,5,9,11,13,15,17,19,21,23



$$T(u,v) = (x,y) = \big(x(u,v),\, y(u,v)\big)$$

$$\vec{a} = \overrightarrow{AB} = \big(\ x(u_0+\Delta u\ ,\ v_0)\ ,\ y(u_0+\Delta u\ ,\ v_0)\ \big) - \big(\ x(u_0,v_0)\ ,\ y(u_0,v_0)\ \big)$$

$$= \big(\ x(u_0+\Delta u\ ,\ v_0) - x(u_0,v_0)\ ,\ y(u_0+\Delta u\ ,\ v_0) - y(u_0,v_0)\ \big)$$

$$= \left(\frac{\partial x}{\partial u}\Delta u, \frac{\partial y}{\partial u}\Delta u\right)\Bigg|_{(u^*,v^0)} \qquad u^* \text{介於} u_0+\Delta u \ and \ u_0 \text{之間} .$$

$$= \Delta u\left(\frac{\partial x}{\partial u}, \frac{\partial y}{\partial u}\right)\Bigg|_{(u^*,v^0)}$$

同理

$$\vec{b} = \overrightarrow{AC} = \left(\frac{\partial x}{\partial v}\Delta v, \frac{\partial y}{\partial v}\Delta v\right)\Bigg|_{(u_0,v^*)}$$

$$= \Delta v\left(\frac{\partial x}{\partial v}, \frac{\partial y}{\partial v}\right)\Bigg|_{(u_0,v^*)}$$

當 Δu 和 Δv 很小時

絕對值

$$R \text{ 的面積} \approx \left|\ \vec{a} \times \vec{b}\ \right| = \left| \begin{vmatrix} \dfrac{\partial x}{\partial u} & \dfrac{\partial x}{\partial v} \\ \dfrac{\partial y}{\partial u} & \dfrac{\partial y}{\partial v} \end{vmatrix} \right| \Delta u \Delta v$$

$$= \left| \begin{vmatrix} \dfrac{\partial x}{\partial u} & \dfrac{\partial x}{\partial v} \\ \dfrac{\partial y}{\partial u} & \dfrac{\partial y}{\partial v} \end{vmatrix} \right| \text{S 的面積.}$$

Definition： The Jacobian of the transf 倍率 T given by $x = g(u,v)$ and

$$y=h(u,v)\ \text{ is }\ \frac{\partial(x,y)}{\partial(u,v)}=\begin{vmatrix}\dfrac{\partial x}{\partial u} & \dfrac{\partial x}{\partial v}\\ \dfrac{\partial y}{\partial u} & \dfrac{\partial y}{\partial v}\end{vmatrix}=(\text{行列式値})$$

註(i) 當 Δu , Δv 夠小時轉換後的面積倍率為 Jacobian 的絕對值

(ii) $\Rightarrow dA=dxdy=\left|\text{Jacobian}\right|\ dudv$

(iii) 若 T 為 Linear Transformation, 則任一區域 S(不須小)經 T 轉換後之區域 R 面積為原面積的 $\left|\text{Jacobian}\right|$ 倍.

Theorem：

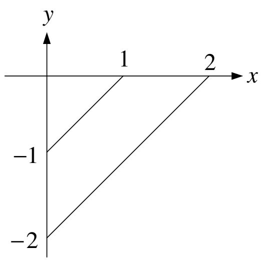
(i) $$\iint_R f(x,y)\,dA=\iint_S f\left(x(u,v),y(u,v)\right)\left|\frac{\partial(x,y)}{\partial(u,v)}\right|dudv$$

(ii) $$\iiint_R f(x,y,z)\,dV=\iiint_v f\left|\frac{\partial(x,y,z)}{\partial(u,v,w)}\right|\ dudvdw$$

註：(1) $\dfrac{\partial(x,y,z)}{\partial(\rho,\theta,\phi)}=-\rho^2\sin\phi$.　　(2) $\dfrac{\partial(x,y)}{\partial(r,\theta)}=r$.

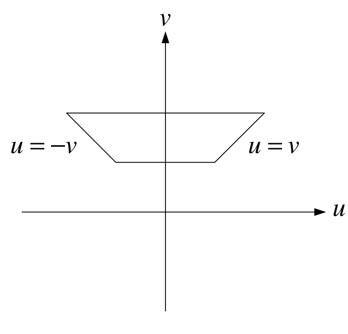
Example 1： $\displaystyle\iint_R e^{\frac{x+y}{x-y}}\ dA$.

$R:$



令 $u=x+y$
$v=x-y$

$$\Rightarrow x=\frac{1}{2}(u+v)$$
$$y=\frac{1}{2}(u-v)$$



$$=\iint_S e^{\frac{u}{v}}\left|\frac{\partial(x,y)}{\partial(u,v)}\right|\ dA=\frac{}{2}\iint_S e^{v}\ dA$$

$$=\frac{1}{2}\int_1^2\int_{-v}^{v}e^{\frac{u}{v}}dudv=\frac{3}{4}\left(e-e^{-1}\right)$$

Example 2： $\iiint\limits_E dV \qquad E: \dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}+\dfrac{z^2}{c^2}=1$

Set $x=au$, $y=bv$, $z=cw$

$$= \iiint\limits_{u^2+v^2+w^2\le 1} abc\ dV = \frac{4}{3}\pi abc$$

Example 3： $\iint\limits_R e^{x+y}\ dA. \qquad R: |x|+|y|\le 1$

Solution：

$$u=x+y \ \ , \ \ v=x-y$$

$$\Rightarrow x=\frac{1}{2}(u-v)\ ,\ y=\frac{1}{2}(u+v)$$

$$\frac{\partial(x,y)}{\partial(u,v)}=\frac{1}{2}$$

$$\Rightarrow \iint\limits_R e^{x+y}\ dA = \iint\limits_S \frac{1}{2}e^u dA$$

$$=\frac{1}{2}\int_{-1}^{1}\int_{-1}^{1} e^u du dv = e-e^{-1}$$